

AET 5420 Quiz 4

Audio Signal Processing

Due: 3pm on April 24, 2024

1 IMPLEMENTING IIR FILTERS IN THE FREQUENCY DOMAIN

IIR filters can be implemented using frequency-domain processing based on a transfer function:

$$Y[z] = H[z] \cdot X[z]$$
$$\text{Where: } H[z] = \frac{b_0 z^{-0} + b_1 z^{-1} + b_2 z^{-2} + \dots}{a_0 z^{-0} + a_1 z^{-1} + a_2 z^{-2} + \dots}$$

To perform frequency-domain filtering, a complex-valued filter, $H[z]$, must be created. This can be done by using the relationship between the Z-transform and the Fourier Transform:

$$z^{-d} = e^{-j \cdot 2\pi \cdot k \cdot d \cdot \frac{1}{N}}$$

Essentially, this substitution should be made in the transfer function for each term. As an example: $z^{-3} = e^{-j \cdot 2\pi \cdot k \cdot 3 \cdot \frac{1}{N}}$. Keep in mind, N is the total number of samples used for the FFT of $X[k]$ and k is an array of integers from $[0, 1, 2, \dots, N-1]$. Therefore, the numerator of $H[k]$ is an array with length N after summing all of the terms. The same is true for the denominator. After determining the transfer function, element-wise multiplication can be used between $H[k]$ and $X[k]$ to calculate $Y[k]$, which is the frequency-domain version of the processed output signal.

1.1 PROBLEM

Create and save a **function** (m-file) in MATLAB that performs frequency-domain IIR filtering:

- Name the function: `freqFilter.m`
- It should have the following input and output variables
 - `[y] = freqFilter(b,a,x)`
 - `b`: array of feedforward coefficients
 - `a`: array of feedback coefficients
 - `x`: input signal
 - `y`: output signal
- Here are some guidelines and hints:
 - The output signal should end up being the same length as the input signal
 - The function should work with any length input signal
 - The function should work with arrays `b` and `a` of arbitrary length
 - It is not necessary for `b` and `a` to be the same length
- My recommendation would be to start this problem with a specific case, then build to the general solution. Examples:
 - `b = [1 -1], a = [1]`
 - `b = [1], a = [1 0.5]`
 - `b = [1 1], a = [1 0.5]`
 - `b = [0.2929, -.5858, .2929], a = [1, 0, 0.1716]`
- The function should perform frequency-domain processing
 - `X = fft(x)`
 - `y = real(ifft(Y))`
 - `N = length(x)`
 - `k` has values 0, 1, 2, ... , N-1
- The function will need to create, `H`, based on `b` and `a`
 - It is recommended to create a numerator, `Hnum`, and denominator, `Hden`
 - $H = \frac{H_{num}}{H_{den}}$
- You may choose to process each frequency bin, `k`, one at a time. Or, you could process the entire spectrum in one command.
- Use the provided test script to verify the performance of the function
 - Experiment with taking the impulse response of the function
 - Plot the results and compare to `freqz(b,a)`
 - Filter white noise and listen to the result

For this problem, you will submit the function file, **freqFilter.m**, and also **testScript.m**.

1.2 SUBMISSION

When completed, put the script and your function in a compressed zip folder. Name the zip file: X_AET5420_Quiz4.zip, where X is your last name. Then email the zip file to: eric.tarr@belmont.edu before 3pm on Wednesday April 24th.